



## *Cordia verbenacea* and secretion of mast cells in different animal species

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### ARTICLE INFO

#### Article history:

Received 22 October 2010

Received in revised form 16 March 2011

Accepted 21 March 2011

Available online 29 March 2011

#### Keywords:

Boraginaceae

*Cordia verbenacea*

Mast cell

Histamine

Inflammation

Allergy

### ABSTRACT

**Ethnopharmacology relevance:** Different plant species from *Cordia* genera are used in folk medicine as anti-inflammatory medication throughout the tropical and subtropical regions of the world. In Brazil, *Cordia verbenacea* is a medicinal plant known as “erva-baleeira”. The alcoholic extracts, decoctions and infusions with leaves of *C. verbenacea* are used in Brazilian traditional medicine for treatment of cough, pneumonia, parasitic diseases and, especially, the inflammatory processes. Anti-inflammatory activity was already demonstrated; however, molecular mechanisms of action are not completely understood. Considering the importance of histamine in early events of inflammation and in allergic diseases, we evaluated the effect of ethanol extract of leaves of *C. verbenacea* on histamine release (*in vitro* and *in vivo* studies) from different types of mast cells induced by chemical agents using several species of rodents. **Materials and Methods:** The extraction and quantification of histamine were performed by using an automatic fluorometric continuous flow system.

**Results:** The extract of *C. verbenacea* (30 µg/ml) reduced the *in vitro* secretion of histamine from rat mast cells induced by ionophore A23187, concanavalin A and compound 48/80, respectively, to  $22.1 \pm 2.2\%$ ,  $24.3 \pm 2.5\%$  and  $21.4 \pm 2.1\%$ . At the same concentration, the extract also inhibited the secretion of histamine from mast cells of guinea pig induced by ionophore A23187 to  $33.3 \pm 2.2\%$ , and mast cells of hamster induced by ionophore A23187 and concanavalin A to  $15.8 \pm 2.5\%$  and  $10.8 \pm 2.6\%$ , respectively. The oral treatment with the extract (300 mg/kg) also inhibited the secretion of histamine induced by A23187 about to  $36.3 \pm 3.2\%$  in rats.

**Conclusions:** *C. verbenacea* inhibits the *in vitro* secretion of histamine from mast cells of different animal species, as well as the secretion of mast cells from animals treated with the extract, which gives not only the proven anti-inflammatory effect of the plant, but also anti-allergic effect, opening new possibilities for future anti-allergic herbal medicine.

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### 1. Introduction

*Cordia verbenacea* DC (Boraginaceae) syn. *Cordia curassavica* (Jacq.) Roem. & Schult is a medicinal plant widely distributed in coastal regions of Brazil, popularly known as “erva-baleeira” (Taroda and Gibbs, 1987). The alcoholic extracts, decoctions and infusions with leaves of *C. verbenacea* are used in Brazilian traditional medicine for treatment of the inflammatory processes, cough, pneumonia and parasitic diseases (Bayeux et al., 2002; Sertié et al., 2005).

Anti-inflammatory activity of *C. verbenacea* has been demonstrated in several different models of inflammation with both leaves extract (Bayeux et al., 2002; Sertié et al., 1988, 1991, 2005) and essential oil (Fernandes et al., 2007; Medeiros et al., 2007; Passos et al., 2007). In this latter oil, the anti-inflammatory property was mainly attributed to  $\alpha$ -humulene and (–)-trans-caryophyllene (Fernandes et al., 2007; Medeiros et al., 2007; Passos et al., 2007). Besides, the crude ethanol extract of leaves of *C. verbenacea* showed remarkable analgesic and antiulcer activity, as well as antioxidant activity in different models of ulcer and pain (Roldão et al., 2008). In addition, both the extract and the essential oil demonstrated to have antibacterial activity (De Carvalho et al., 2004; Meccia et al., 2009; Michielin et al., 2009).

In early events of inflammation, histamine plays an important role, as well as in the events related to allergic reactions such as asthma, rhinitis, food allergy and dermatitis (Metcalf and Boyce, 2006). Histamine is released from mast cells, as well as other

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pro-inflammatory mediators, after the chemical or immunological stimulation of these cells (Shin et al., 2004; Metcalfe and Boyce, 2006).

However, the secretion of mast cells varies depending on the origin (mucosa or connective tissue) and, also, there is variation in sensitivity to the effects of histamine in different animal species (Ferreira et al., 1996; Gomes et al., 1994). Since the inhibition of the effects of histamine and mast cells secretion is the main therapeutic approach for the treatment of allergic reactions, the latter differences ultimately influence the effects of the drugs currently used in therapy of allergic diseases and reduce the effectiveness of treatment (Gomes et al., 1994).

In this way, the use of natural products as source of new active compounds (Shen et al., 2008) for complementary therapy is an important therapeutic strategy in order to obtain a better management of allergic diseases.

Thus, the objective of this study was to investigate the direct influence of *C. verbenacea* in the secretion of histamine from mast cells induced by different secretory agents *in vitro* and to compare the effect of the extract with the effects of known inhibitors of mast cell secretion. Moreover, this study also evaluated the *in vivo* influence of *C. verbenacea* in the secretion of histamine from mast cells of animals pretreated with the extract.

## 2. Materials and methods

### 2.1. Plant material

After collected in Ubatuba, State of São Paulo (Brazil) on January 2006, plants were identified as *Cordia verbenacea* DC. (Boraginaceae) by comparing with a voucher specimen (#28117) deposited in the Herbarium SJRP of the IBILCE/UNESP (São José do Rio Preto, State of São Paulo, Brazil).

### 2.2. Preparation of the extract

#### 2.2.1. Ethanol extraction

Leaves of *C. verbenacea* (2760g) were dried at 50 °C for 72 h, powdered and weighed, yielding 640 g (dry weight/wet weight = 23.2%). Then, the dried leaves were packed in glass jars and subjected to absolute ethanol extraction (640 g of plant with 1 l of absolute ethanol) by cold maceration. The container was kept in the dark in cold (4 °C for 72 h). Then, the extract was filtered and the rest of the plant was extracted three times with absolute ethanol up to exhaust the liquid material. After extraction, the total volume was concentrated in rotary evaporator at 45 °C up to 20% of original volume.

#### 2.2.2. Elimination of chlorophyll

The concentrated extract was subjected to extraction of chlorophyll as described elsewhere (Roldão et al., 2008). Briefly, distilled water was added in drops (4 ml/min) under stirring until obtaining an ethanolic solution of 70% (ethanol:water, 7:3).

Subsequently, the extract was cooled at 4 °C for 24 h and filtered through celite in vacuum system, obtaining a precipitate rich in chlorophyll and chlorophyll-free ethanol extract. Finally, for the elimination of the ethanol, the extract free of chlorophyll was concentrated in a rotary evaporator and the precipitate was obtained. The total extraction yield was 3.6% (weight of final extract/weight of dry plant) for this extraction protocol.

### 2.3. Reagents

Compound 48/80, bovine serum albumin (BSA), collagenase type IA, ionophore A23187, concanavalin A (Con A), theophylline,

sodium cromoglicate and quercetin were purchased from Sigma Chemical Co (St. Louis, MO, USA).

### 2.4. Animals

Male Wistar rats (180–200 g), hamsters (80–120 g) and Dunkin–Hartley Guinea pigs (280–350 g) were housed in standard environmental conditions (21 °C, 60–70% humidity) with 12 h light/dark cycle.

This research was conducted in accordance with the internationally accepted principles for laboratory animal use and care, and all the experimental protocols were approved by the Commission of Ethics in Animal Experimentation of the Institute of Biosciences of the University of São Paulo State (UNESP, Botucatu, São Paulo, Brazil).

### 2.5. Obtaining cell suspensions

#### 2.5.1. Suspension of mast cells of rat peritoneum

Thirty milliliters of Tyrode's solution (137 mM NaCl, 2.7 mM KCl, 1 mM CaCl<sub>2</sub>, 0.4 mM NaHPO<sub>4</sub>, 12 mM NaHCO<sub>3</sub> and 5.6 mM glucose), pH 7.4, containing heparin (5 units/ml), were injected into the peritoneal cavity of the rats. The cells were recovered by filtration of abdominal fluid with a moistened gaze, centrifuged (150 × g for 4 min at room temperature) and washed twice with Tyrode solution free of heparin. Finally, cells were resuspended in Tyrode's solution (approximately 4 × 10<sup>5</sup> cells/ml, final concentration) for the assessment of histamine secretion.

#### 2.5.2. Suspensions of guinea pig lung mast cells and mast cells of hamster cheek pouch

Cell suspensions were obtained by enzymatic dispersion with collagenase according to Ferreira et al. (1996). Briefly, tissues were cut into small fragments (1–2 mm<sup>2</sup>) and incubated (90 min, 37 °C) in Tyrode's solution containing bovine serum albumin (BSA, 1 mg/ml), type IA collagenase (160 units/ml) and HEPES 10 mM. This material was filtered and centrifuged (150 × g for 4 min at 4 °C). The pellet was then washed with HEPES-Tyrode containing BSA (0.5 mg/ml) and Tyrode's solution. Finally, the cells were resuspended in Tyrode's solution for the assessment of histamine secretion.

### 2.6. In vitro treatments

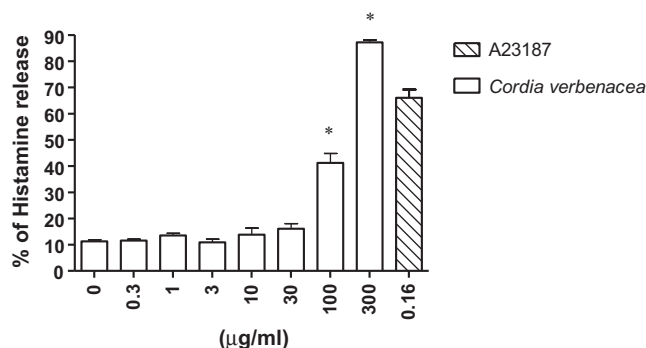
Suspensions of mast cell were stabilized for 5 min at 37 °C in a water bath. Cells were then incubated for 10 min at 37 °C with 0–30 µg/ml of the *C. verbenacea* extract, vehicle (33.3% ethanol) or drugs that inhibit secretion of histamine (quercetin, cromolyn sodium and theophylline). Subsequently, compound 48/80 (0.5 µg/ml), concanavalin A (100 µg/ml) or ionophore A23187 (0.3 µM) were added and, after 10 min, the secretion was interrupted with cold Tyrode's solution.

### 2.7. In vivo treatments

To evaluate a possible systemic effect of *C. verbenacea* on the secretion of mast cells, rats were treated orally with 300 mg/kg of the extract (diluted in ethanol:water, 1:3), vehicle or water, once daily for 2 days. All the animals were killed 6 h after the last administration and a cell suspension containing mast cells was obtained from rat peritoneum as described before (see Section 2.5.1)

### 2.8. Determination of histamine release

After treatments, all suspensions were centrifuged at 150 × g and 4 °C for 5 min. Supernatants were transferred to clean



**Fig. 1.** Histamine release in mast cells from rat peritoneum treated *in vitro* with the hydro-ethanolic extract 70% of leaves of *Cordia verbenacea* ( $n=7$ ). \* $p < 0.05$  versus control group and A23187 group.

tubes and pellets were resuspended in 1.5 ml of Tyrode solution. Perchloric acid (0.4N, final concentration) was added to both supernatants and resuspended pellets. Histamine content was determined by automatic fluorometry (Shore et al., 1959). Finally, histamine release was calculated as the difference between the content of histamine in supernatants and resuspended pellets.

### 2.9. Statistical analysis

Data were expressed as mean  $\pm$  SEM. Student's *t* test for paired samples and one-way analysis of variance (ANOVA) was performed followed by the *post hoc* test Tukey–Kramer when appropriated. The level of significance was set at  $p < 0.05$ .

## 3. Results

### 3.1. Concentration-dependent effect of *C. verbenacea* extract on histamine release

Both vehicle and extract of *C. verbenacea* (with concentrations  $\leq 30$  µg/ml) did not alter either the spontaneous release of histamine or not the total content of histamine of mast cells suspension of rat peritoneum (Fig. 1). Only when higher concentrations (100 and 300 µg/ml) of the extract were used, a significant increase ( $p < 0.05$ ) in spontaneous release of histamine was detected (Fig. 1). Therefore, these concentrations were excluded from the study.

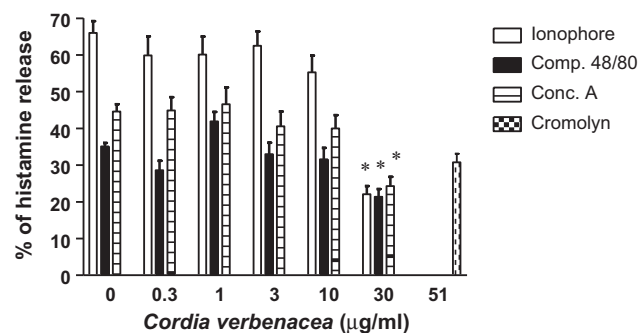
Likewise, treatment of mast cells suspensions from guinea pig lung and hamster cheek pouch with 30 µg/ml of *C. verbenacea* extract did not vary histamine release when compared to vehicle group ( $6.3 \pm 0.7$  and  $11.3 \pm 1.1\%$  for spontaneous release and  $8.8 \pm 1.3$  and  $13.8 \pm 1.0\%$  for treated groups of the two species, respectively,  $p > 0.05$ ).

### 3.2. Inhibition by *C. verbenacea* extract of induced histamine release in rat peritoneal mast cells.

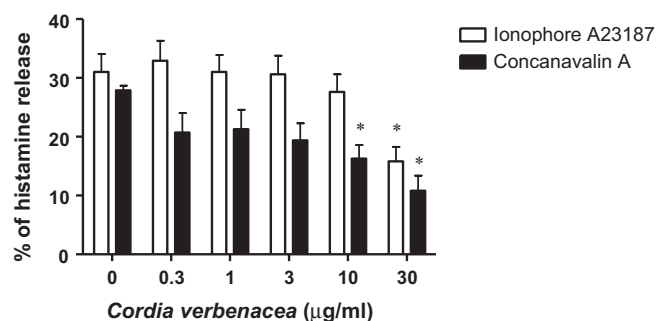
*C. verbenacea* extract (30 µg/ml) was able to significantly ( $p < 0.05$ ) inhibit histamine release in suspensions of mast cells from rat peritoneum, induced by compound 48/80, ionophore A23187 and concanavalin A (Fig. 2).

### 3.3. Inhibition by *C. verbenacea* extract of induced histamine release in mast cells from guinea pig lung

The *C. verbenacea* extract (30 µg/ml) caused reduction of secretion of histamine from mast cells of guinea pig lung induced by



**Fig. 2.** Inhibition by hydro-ethanolic extract 70% of leaves of *C. verbenacea* of induced histamine release (by ionophore A23187 0.3 µM, 0.5 µg/ml of compound 48/80 or 100 µg/µl of concanavalin A) in suspensions of rat peritoneal mast cells ( $n=8-10$ ). \* $p < 0.05$  versus control group and cromolyn sodium.



**Fig. 3.** Inhibition by hydro-ethanolic extract 70% of leaves of *C. verbenacea* of induced histamine release (by ionophore A23187 0.3 µM or 100 µg/ml of concanavalin A) in suspensions of mast cells from hamster cheek pouch ( $n=6-8$ ). \* $p < 0.05$  versus control group.

ionophore A23187 (0.3 µM) to  $33.3 \pm 2.2\%$  when compared to control group (without extract)  $48.2 \pm 2.0\%$  ( $p < 0.05$ ) ( $n=7$ ). The *C. verbenacea* extract in lower concentrations (0.3, 1, 3 and 10 µg/ml) no significant differences were found on histamine release, respectively:  $44.0 \pm 3.4\%$ ;  $45.1 \pm 3.7\%$ ;  $43.3 \pm 3.0\%$ ;  $42.2 \pm 2.9\%$  ( $p > 0.05$ ) ( $n=7$ ).

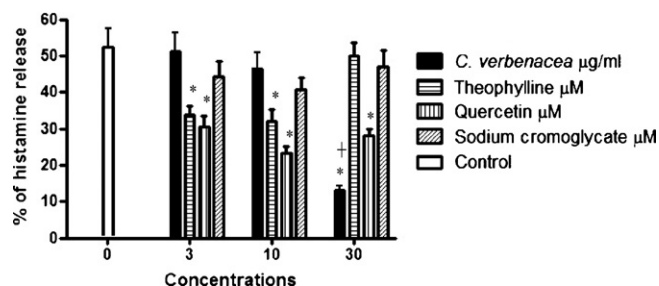
### 3.4. Inhibition by *C. verbenacea* extract of induced histamine release in mast cells from hamster cheek pouch

The leaf extract of *C. verbenacea* (concentrations of 10 and 30 µg/ml) significantly ( $p < 0.05$ ) inhibited histamine release induced by ionophore A23187 or concanavalin A in mast cells from hamster cheek pouch (Fig. 3).

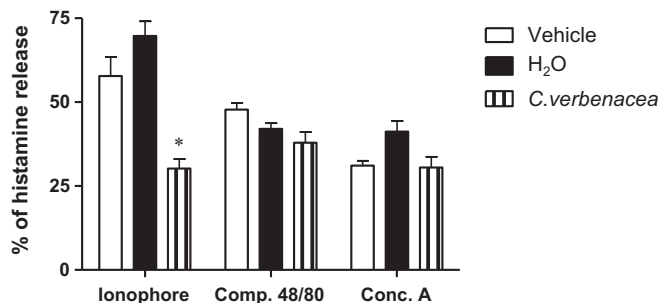
### 3.5. Comparison between classical inhibitors of the secretion of mast cells and the effect of the *C. verbenacea* extract in mast cells suspensions of rat peritoneum

When mast cells suspensions of rat peritoneum were exposed to quercetin and theophylline, a significant ( $p < 0.05$ ) inhibition of histamine release induced by ionophore A23187 was detected with concentrations  $\geq 3$  µM (Fig. 4). For lower concentrations no significant differences were found (data not shown). In addition, incubation with cromoglycate only provoked a detectable inhibition when a concentration of 100 µM was used (induced histamine release was reduced to  $30.8 \pm 2.3\%$ ,  $p < 0.05$ ).

In comparison, the inhibitory effect of 30 µg/ml of the *C. verbenacea* on histamine release was higher than that observed with 30 µM of theophylline or quercetin ( $p < 0.05$ ) (Fig. 4), pointing to a more effective action of the extract.



**Fig. 4.** Effect of hydro-ethanolic extract 70% of leaves of *C. verbenacea*, sodium cromoglycate, quercetin and theophylline in histamine release induced by 0.3 µM of ionophore A23187 from mast cells of rat peritoneum ( $n = 7-9$ ). \* $p < 0.05$  versus control; † $p < 0.05$  versus all groups.



**Fig. 5.** Effect hydro-ethanolic extract 70% of leaves of *C. verbenacea* (300 mg/kg) on histamine release induced by ionophore A23187 (0.3 µM), compound 48/80 (0.5 µg/ml) and concanavalin A (100 µg/ml) in mast cells of rat peritoneum ( $n = 7$ ). \* $p < 0.05$  versus vehicle and H<sub>2</sub>O groups treated with the same inductor.

### 3.6. Effect of oral treatment with extract of *C. verbenacea* (300 mg/kg) on histamine release in mast cells of rat peritoneum

The oral treatment of rats with extract of *C. verbenacea* inhibited histamine release induced by ionophore A23187 (0.3 µM), ( $p < 0.05$ ) but did not inhibit histamine release induced by compound 48/80 or concanavalin A ( $p > 0.05$ ) (Fig. 5).

## 4. Discussion

The secretion of mast cells and histamine release is the start point to the genesis of the allergic processes. The actions of histamine, as well as other pro-inflammatory substances released during secretion (prostaglandins, chemokins and leukotrienes among others), are responsible for the symptoms, hence inhibiting the secretion mast cells results in better therapeutic response to treatment of allergy.

However, mast cells are phenotypically heterogeneous and exhibit variable degrees of sensitivity against inducers or inhibitors of secretion (Marone et al., 1997). This fact represents a challenge in the pursuit of more effective anti-allergic drugs. Thus, in this study we evaluated the effect of leaf extract of *C. verbenacea* in mast cells obtained from different species and stimulated by different chemical agents (compound 48/80, concanavalin A and ionophore A23187) to induce secretion by distinct mechanisms, although none of them acts in a cytotoxic manner (Gomes et al., 1994).

Our results demonstrated, for the first time, that the leaf extract of *C. verbenacea* shows wide *in vitro* inhibitory activity of histamine release induced by different agents and in different tissues and animal species (guinea pig, hamster and rat).

The compound 48/80 acts directly on the mast cell by binding to G protein-coupled receptors, this interaction leads, via phospholipase C and D, the increased activity of protein tyrosine kinase C

and increased intracellular calcium flux, occurring then cell degranulation (Mousli et al., 1990; Palomaki and Laitinen, 2006). The concanavalin A, in turn induces the release of histamine interacting with residual sugar and metals on the surface of mast cells, however, the release is not primarily to the immune system, meaning that the presence of immunoglobulin E in conjunction with mast cells is not essential for the secretory stimulus occurs (Lopes et al., 2005).

Moreover, the ionophore A23187 plays a role as a carrier of calcium from the extracellular milieu to intracellular as well as promotes the release of calcium from organelles (Gomes et al., 1994). This increase in calcium activates protein kinase C and induces phosphorylation of myosin light chain in the cytoskeleton of mast cells initiate the process of exocytosis (Ludowyke et al., 1996).

Thus, a change in the availability of intracellular calcium could be involved in the molecular mechanism of action of the *C. verbenacea* extract, since all stimulating agents used in this study depend on the presence of calcium to trigger the release of histamine.

It is important to highlight that the vehicle (ethanol/water) used to dilute the extract was assessed to verify possible interferences in the secretion of mast cells in our models. Although ethanol is able to modify the secretion in this type of cells, it is a concentration-dependent way (Ruiz and Gomes, 2000), and no effect was detected on histamine release in our experimental conditions.

In order to characterize the inhibitory effect of the *C. verbenacea* extract (30 µg/ml), classical inhibitors of histamine release sodium cromoglycate (3–30 µM or 1.54–15.4 µg/ml), quercetin (3–30 µM or 1.0–10 µg/ml) and theophylline (3–30 µM or 0.54 µg/ml 5.4 µg/ml) were used for comparison. As a result, the inhibition of histamine secretion by the extract was more potent than that demonstrated by the cromoglycate (Fig. 4). In our study, cromoglycate was able to inhibit histamine release only when a concentration of 100 µM (or 51 µg/ml) was used (Fig. 2).

Moreover, considering that the concentration of active compounds of the extract (directly responsible for the inhibition) is below 30 µg/ml, these compounds could have similar (or still better) potency than quercetin or theophylline. However, studies with active fractions isolated must be carried out to verify this hypothesis.

Sodium cromoglycate or cromolyn is a chromone derivative of the medicinal plant *Ammi visnaga* (Apiaceae) (Mitchell, 2006) and a classical inhibitor of the mast secretion used experimentally. Although it has been used for nearly 30 years as anti-allergic therapy, its molecular mechanism is not yet fully understood (Mitchell, 2006). In rat peritoneum mast cells, this drug may inhibit the secretion through immunological or non-immunological stimuli. However, Shin et al. (2004) suggested that cromolyn indirectly affects the secretion of mast cells by blocking calcium mobilization.

In turn, theophylline shows a remarkable bronchodilator effect, immunomodulatory activity as well as an inhibitory action on the secretion of mast cells (Page, 1999). For these reasons, this drug has been largely used to treat asthma for almost 50 years (Page, 1999). Such effects could be related to a nonspecific inhibition of phosphodiesterase enzyme and an accumulation of intracellular cAMP, besides to blocking adenosine receptors A1 and A2 (Coward et al., 1998; Weston et al., 1997).

Considering that an extract was used in our study, it is very complex to predict the exact mechanisms of action of the principles involved in the effect, but it is probable that more than one active compound included in the extract is involved in the inhibition, acting in a synergistic way and providing greater efficiency in relation to the inhibitors used for comparison.

In our experiments, *C. verbenacea* also showed inhibitory effect on the secretion of mast cells in animals treated orally with the



extract (Fig. 5). However, the results achieved were different of those observed with the *in vitro* experiments (Figs. 1–4). These differences may be due to possible changes in the bioavailability of the active compounds of the extract under pharmacokinetic process. The fact that pharmacokinetic characteristics of the extract influence the concentrations of the active compounds reaching the mast cells may be also a hypothesis to explain the histamine release induced by ionophore A23187 produced by oral administration of the plant extract, but not by other agents as compound 48/48 or concanavaline A. Thus, new *in vivo* experiments must be carried out with different times of treatment and doses to ensure an adequate intracellular concentration promoting the same effects observed *in vitro*.

Our results are also supported by previous *in vivo* studies with oral administration showing pronounced anti-allergic and anti-inflammatory effects with essential oil and isolated compounds from the oil of *C. verbenacea* in several models of inflammation and allergy (Sertié et al., 1988; Fernandes et al., 2007; Medeiros et al., 2007; Passos et al., 2007). For example, the essential oil of *C. verbenacea* produced a marked reduction of paw edema induced by intraplantar injection of different mediators, including histamine, bee venom or ovalbumin in previously sensitized animals (Passos et al., 2007). Other parameters of anti-allergic activity *in vivo* were also observed in model animals previously sensitized and challenged with ovalbumin antigen, such as reducing the accumulation of pleural exudate, the total content of cells, and migration of mononuclear cells and eosinophils in the pleural cavity, suggesting a possible use of compounds of *C. verbenacea* for the treatment of allergic diseases (Passos et al., 2007).

However, to date no one had evaluated the effect of *C. verbenacea* on histamine release from mast cells induced by different mechanisms.

The possible compounds of the extract responsible for the potent inhibitory effect of *C. verbenacea* include some candidates. Studies with the phytochemical characterization of *C. verbenacea* identified the presence of tannins, mono and sesquiterpenes (trans-caryophyllene, alpha-pinene and alpha-humulene), naphthoquinones, cordiaquinones and flavonoids including quercetin artemetin and hidroartemetin (De Carvalho et al., 2004; Sertié et al., 1990, 2005). Also, polyphenolic compounds flavonoids including kaempferol and quercetin have inhibitory effect on the release of histamine, leukotrienes and cytokines from mast cells of rodents and humans (Pearce, 1986; Duraisamy et al., 2005). More recently our research group identified the presence of flavonoids, xanthenes, triterpenoid glycosides, steroidal aglycone, saponins, alkaloids and phenolic compounds in crude ethanol extract of leaves of *C. verbenacea* (Roldão et al., 2008). The extract used in the present study was the same used by Roldão et al. (2008).

These compounds exhibit activities in relevant organic processes. Xanthenes, for example, show antioxidant, anti-inflammatory, anti-ulcer, antiplatelet and antihypertensive activities (Librowski et al., 2005). Saponins appear to exert an anti-inflammatory action in ulcers and triterpenoids demonstrated anti-inflammatory effect in rat paw edema model (Jung et al., 2005). Flavonoids have a wide variety of biological activities highlighting the antioxidant action, in the addition to the already mentioned inhibition of mast cells (Matsuda et al., 2003; Morikawa et al., 2006.).

Other studies also proved the potent anti-inflammatory activity of (–)-trans-caryophyllene  $\alpha$ -humulene compounds isolated from the leaf essential oil of *C. verbenacea* (Medeiros et al., 2007).

The inhibitory activity of leaf extracts of *C. verbenacea* in the secretion of histamine from mast cells of different tissues and species showed in the present study indicates the broad effectiveness of the active principles of *C. verbenacea*, giving an additional importance on its anti-inflammatory activity. Such properties, anti-

inflammatory and anti-allergic, already support this plant as a future phytotherapeutic product for use in allergic diseases and other diseases related to mast cell secretion.

## 5. Conclusions

The proof of the direct anti-allergic effect of *C. verbenacea* on histamine release from heterogeneous mast cells and, still, for different animal species, never was showed until now. The results of the present work already support the therapeutic potential of *C. verbenacea* in allergic diseases involving histamine.

## Acknowledgments

The authors especially thank Dr. Neusa Taroda Ranga by identifying the plant. This study was supported by grants from CAPES. The research was conducted under the auspices of CYTED (Subprogram X, Project X PIGASTRIN 10). D.M. Costa de Oliveira thanks FAPESPA for her grant.

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